SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Technical Presentation

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 3 – Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	Sree B	**Reg. No.:**	192524023
Section / Batch:	AI and DS / 2025	**Date:**	15/07/2026

Code of Conduct

I Sree B ____________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sree B ____________

Instructions

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation.
- Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

Section / Topic	Focus	Q Nos	Marks
Data Structure	Real-Time Applications	1	100
	GRAND TOTAL:	**100 Marks**	

Annexure A – Technical Presentation

A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency. (100 Marks)

(OR)

A music streaming application maintains a playlist where songs can be added, removed, or reordered dynamically. Recommend a suitable data structure and justify based on flexibility and efficiency. Prepare a technical Presentation. (100 Marks)

(OR)

A metro rail network maps stations and routes to determine reachable stations from a source station. Suggest a suitable data structure for representation and justify your answer. (100 Mark)

(OR)

A calculator application evaluates arithmetic expressions entered by users. Recommend a suitable data structure and justify based on expression processing efficiency. (100 Marks)

(OR)

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	20	Demonstrates deep and accurate understanding of data structure with insights and connections	Shows clear understanding with minor gaps	Basic understanding; some misconceptions present	Limited or incorrect understanding
Content Organization	30	Logical, well-structured, smooth flow; strong introduction and conclusion	Mostly organized with clear flow	Some organization but lacks clarity or transitions	Disorganized, difficult to follow
Technical Depth	20	Includes algorithms, complexity analysis, and advanced insights	Includes algorithm and basic complexity analysis	Limited technical details; missing depth	Minimal or no technical content
PowerPoint Slides	20	Excellent design, clear visuals, enhances understanding	Good slides with minor issues	Basic slides; somewhat cluttered or unclear	Poor slides or no visual support
Communication Skills	10	Clear, confident, engaging delivery; excellent articulation	Clear delivery with minor hesitation	Some hesitation; unclear in parts	Poor communication; difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: ______________	Signature: ______________	Signature: ______________
Date: _________________	Date: _________________	Date: _________________

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Cafeteria Token System Using Queue ADT

GUDIDED BY:
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PRESENTED BY:
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Introduction:

- A cafeteria serves many customers during busy hours.
- To manage the crowd, each customer is given a unique token when they arrive.
- Customers are served in the same order in which they receive their tokens.
- This follows the **First In, First Out (FIFO)** principle.
- The **Queue Abstract Data Type (ADT)** is the most suitable data structure for implementing this system.
- It ensures fair, organized, and efficient service for all customers.

Problem Statement:

- A cafeteria receives many customers during peak hours.
- Customers should be served strictly in the order they arrive.
- Manual management can lead to confusion, skipped customers, and unfair service.
- An efficient system is needed to organize the waiting line.
- The solution should ensure **First Come, First Served (FIFO)**, improve efficiency, and provide equal service to every customer.

Suitable Abstract Data Type (ADT): Queue

- A **Queue** is the most suitable ADT for a cafeteria token system.
- It follows the **FIFO (First In, First Out)** principle.
- The customer who receives the token first is served first.
- New customers are added to the **rear** of the queue (**Enqueue**).
- Customers are served and removed from the **front** of the queue (**Dequeue**).
- This ensures **fair, organized, and efficient** customer service.

Queue Operations:

1. Enqueue

- Adds a new customer (token) to the **rear** of the queue.

2. Dequeue

- Removes and serves the customer at the **front** of the queue.

3. Front (Peek)

- Displays the next customer to be served without removing them.

4. IsEmpty()

- Checks whether the queue is empty.

5. IsFull() ***(for array-based queues)***

- Checks whether the queue has reached its maximum capacity.

Working of Cafeteria Token System:

- A customer enters the cafeteria and receives a unique token.
- The customer's token is added to the rear of the queue (Enqueue).
- Customers wait in the order of their arrival.
- When the counter is available, the customer at the front of the queue is served first (Dequeue).
- After service, the next customer in the queue moves to the front.
- This process continues until all customers have been served.

Advantages of Queue ADT:

- Ensures Fairness: Customers are served in First In, First Out (FIFO) order.
- Easy to Implement: Simple operations such as Enqueue and Dequeue.
- Organized Customer Management: Maintains a systematic waiting line.
- Reduces Waiting Confusion: Prevents customers from skipping the queue.
- Improves Efficiency: Speeds up the service process during busy hours.
- Scalable: Can handle both small and large numbers of customers.
- Widely Applicable: Used in cafeterias, banks, hospitals, railway stations, ticket booking systems, and call centers.

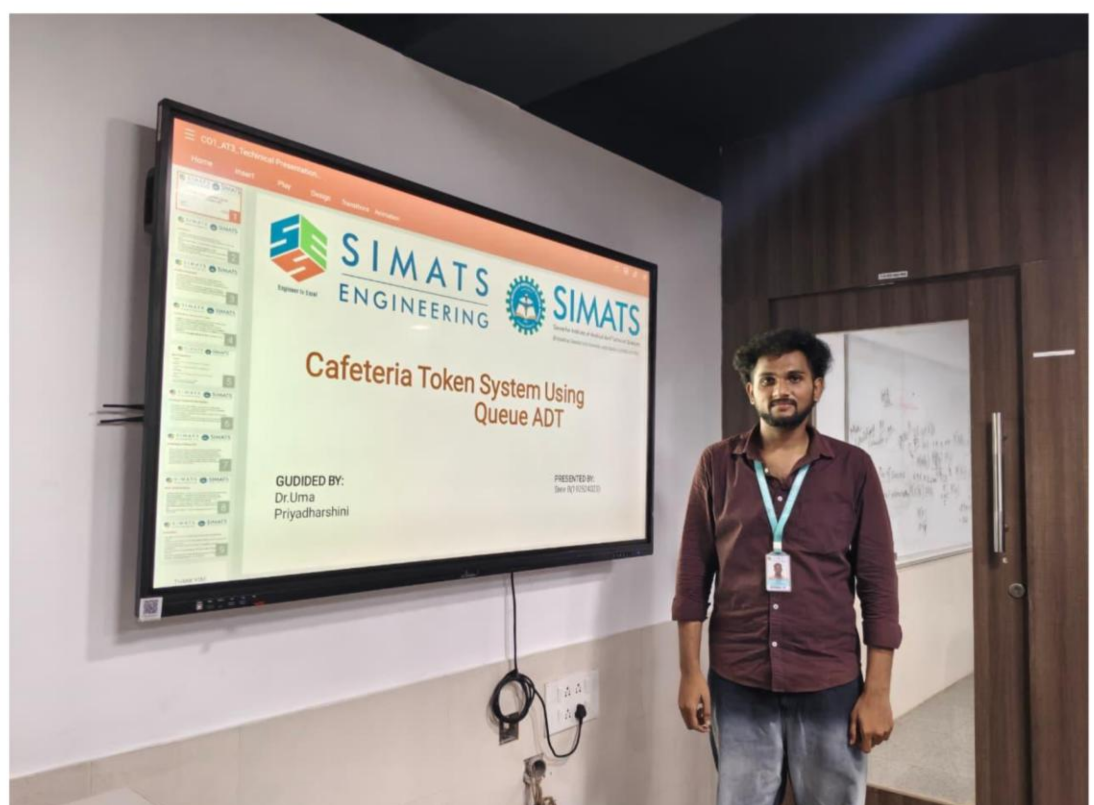
Real-Life Applications:

- **Cafeteria Token System** – Customers are served in the order they receive tokens.
- **Bank Token System** – Customers wait for their turn at service counters.
- **Hospital Registration** – Patients are attended based on their arrival order.
- **Railway and Bus Ticket Booking** – Passengers are served in a queue.
- **Printer Queue** – Print jobs are processed one after another.
- **Call Centers** – Customer calls are handled in the order they are received.
- **Supermarkets and Billing Counters** – Shoppers are billed in the order they join the line.
- **Online Customer Support** – Support requests are processed in the sequence they are received.

Conclusion:

- The Queue ADT is the most suitable data structure for a cafeteria token system.
- It follows the FIFO (First In, First Out) principle, ensuring customers are served in the order they arrive.
- This approach provides fairness, efficiency, and organized customer management.
- Queue operations such as Enqueue and Dequeue make the system simple and effective.
- Queue ADT is widely used in many real-life applications, including banks, hospitals, ticket booking systems, and call centers.

THANK YOU!



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Cafeteria Token System Using
Queue ADT
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